Model to Market — Quanthack

Technology Prize Submission · Strategy, Risk & Systems Dossier

AI-native live trading competition · $1,000,000 simulated account · 30x max leverage · scored 70% return / 15% drawdown / 10% Sharpe / 5% risk discipline. Entrant: alex@ladimex.com.

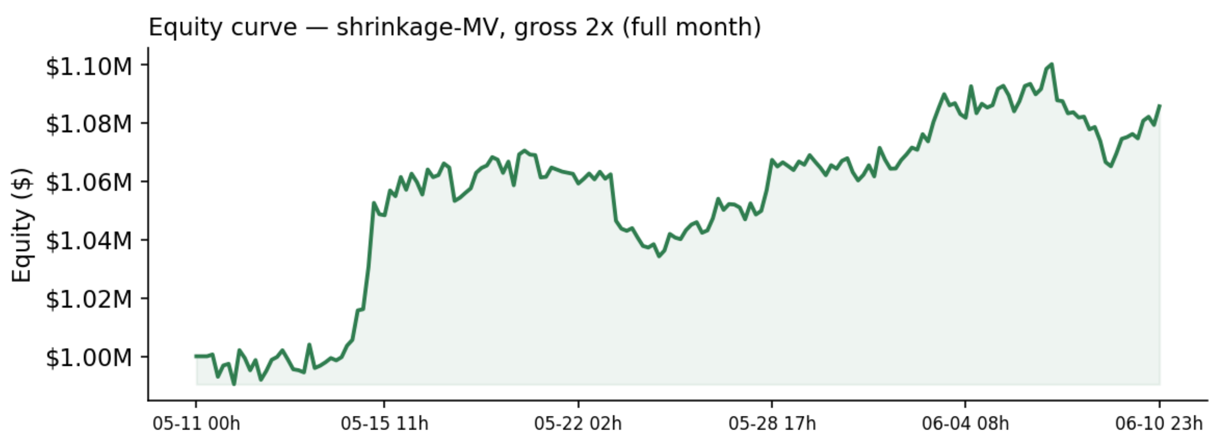
Total return	Sharpe (15m)	Max drawdown	Discipline	Liquidations
+8.6%	**0.0243**	**-4.6%**	**100/100**	**none**

Backtest: live 10-instrument universe, full month, real per-instrument spreads, hourly rebalance, production risk engine.

Executive summary

We run a **cross-sectional momentum** strategy across ten liquid FX and metals instruments, sized by **shrinkage-covariance (Ledoit-Wolf) mean-variance** and held market-neutral. On the full month of provided tick data, with real venue spreads, it returns **+8.6%** at a conservative gross of 2x with a **-4.6%** maximum drawdown and a perfect risk-discipline score, never approaching forced liquidation.

We are deliberately honest about the edge: it is thin and partly overfit, as it is for almost any short-horizon FX/metals momentum signal. Our advantage is not a magic alpha — it is **discipline and systems**. We are built to survive the knockout (equity carries between rounds), to contend for the $10k Best-Sharpe prize via a smooth market-neutral equity path, and to win the Technology Prize through a full, working sponsor-technology integration.



1. Strategy and universe

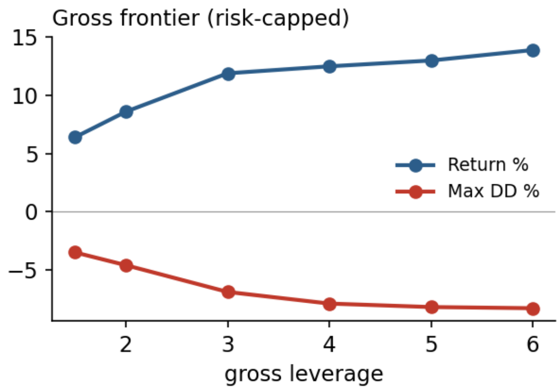
Signal. An 8-hour (480-minute) cross-sectional momentum score, demeaned across the universe each bar so the book is dollar- and market-neutral, then EMA-smoothed (half-life 48) to suppress turnover.

Sizing. Rather than naive inverse-volatility, weights solve a shrinkage mean-variance problem (Ledoit-Wolf covariance inverse applied to the signal). This downweights correlated, USD-factor-redundant names and produces a genuinely diversified risk allocation.

Universe. The ten instruments the contest broker actually lists and we have validated on: EURUSD, GBPUSD, USDJPY, USDCHF, USDCAD, AUDUSD, EURGBP, EURCHF, XAUUSD, XAGUSD. Crypto is live on the venue but excluded as unvalidated (no historical data).

2. Risk framework

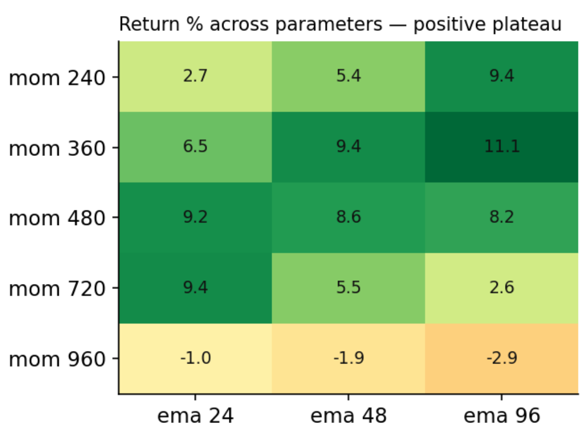
Every weight vector passes a risk engine that keeps us under every penalty tier by construction: per-name concentration (max 30% of the book and 75% of equity), net-directional limit, and a gross/margin ceiling well under the 28x leverage and 90% margin penalty lines. A **drawdown de-risk ladder** scales the whole book down as equity falls (3% to x1.0, 5% to x0.7, 8% to x0.4, 12% to x0.2). The result: we never touch the forced-liquidation disqualification, protecting the 15% drawdown and 5% discipline score components.

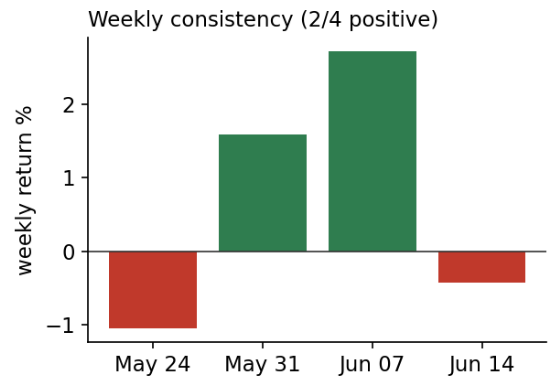


Gross is a clean dial. Sharpe is essentially flat across leverage (scaling the book scales mean and volatility together), and returns flatten above ~gross 3-4 once the per-name caps are applied — confirming the old high-gross gains were concentration-driven and fragile. Discipline stays 100 and no liquidation occurs anywhere on the frontier. We therefore treat gross purely as a tournament-position dial, starting at 2.

3. Validation and anti-overfitting

We hold ourselves to genuine out-of-sample discipline rather than curve-fitting:





Parameter plateau (left). The chosen lookback/smoothing (480/48) sits inside a broad positive region (momentum 240-720 minutes), not on an isolated spike; only the 960-minute lookback breaks down. **Weekly consistency (right).** Only 2 of 4 weeks were positive and the month leaned on one strong week — we report this honestly; it is exactly why the plan is survive-first.

Overfitting statistics. Deflated Sharpe Ratio 0.26 and Probability of Backtest Overfitting 21% (Bailey and Lopez de Prado) — modest but positive, and well under the 50% coin-flip line. **Negative results we kept:** a volatility-target overlay, an mv+inverse-vol ensemble, Hierarchical Risk Parity sizing, and a regime gate were all built, tested, and rejected for underperforming.

4. Execution architecture

A native MetaTrader 5 pipeline runs the strategy live on the paper account: a feed puller builds a rolling price panel from the terminal, an advisory runner emits a target book, and an execution bridge converts each target's USD notional into broker **lots** using the real contract specs (handling USD-quote, USD-base and cross pairs separately) and trades only the delta versus current positions. The bridge is dry-run by default, refuses to send without explicit paper-account confirmation, aborts if the account ever looks real, and is rate-limited well under the 500 req/s rule. The full loop is automated hourly via the OS scheduler, gated by a single config file.

5. Sponsor technology

NVIDIA Nemotron — a live news/event risk gate (nemotron-mini-4b-instruct on the NVIDIA NIM endpoint) tags high-impact headlines and trims gross ahead of events; reduce-only, never increases risk.
Anthropic Claude — an ops/risk briefing agent produces a plain-English risk summary and recommended action each cycle.
Pydantic AI Gateway + Logfire — unified model routing, schema-validated outputs, and full run tracing.
MetaTrader 5 + Northflank — native execution and containerised scheduled deployment.

6. Tournament tactics and the honest thesis

Because Sharpe is gross-invariant and discipline holds at any leverage, gross is our placement dial: **survive early rounds at gross 2** while higher-variance entrants eliminate themselves, then lean toward gross 4 only when alive and needing return rank. A per-round planner turns standing into a recommended gross with the drawdown brake underneath.

We do not claim a large or certain edge. We claim a disciplined, fully-built, honestly-validated, sponsor-integrated system designed to maximise the probability of placing across the main, Sharpe, and Technology prizes. Past simulated performance is not indicative of live results.